PRACTICAL 7. Sampling and Sampling Distributions

A.) Aim:- To conduct a survey and collect data from a sample population using proper sampling techniques.

B.) Aim:- To use the Central Limit Theorem(CLT) to analyze the sampling distribution of a sample mean and estimate population parameters.

Code:-

```
import numpy as np
import matplotlib.pyplot as plt

#Population data
population=np.array([
    2,3,4,2,5,3,4,6,3,5,
    2,4,3,5,4,6,3,4,5,2,
    3,4,5,3,6,4,2,5,4,3,
    5,6,3,4,2,5,3,4,6,5,
    2,3,4,5,3,6,4,2,5,4
])

#Population_Mean
population_mean=np.mean(population)
print("Population Mean:",population_mean)

#Number of Samples
num_samples=1000

#Sample Size
sample_size=10

#Store sample means
sample_means=[]

#Generate samples
for i in range(num_samples):
    sample=np.random.choice(population,
                            size=sample_size)
    sample_mean=np.mean(sample)
    sample_means.append(sample_mean)
```

```
#Mean of sampling distribution
sampling_mean=np.mean(sample_means)
print("Mean of Sampling Distribution:",sampling_mean)

#Plot sampling distribution
plt.hist(sample_means,bins=20,edgecolor="black")
plt.xlabel("Sample Mean")
plt.ylabel("Frequency")
plt.title("Sampling Distribution of Sample Mean")
plt.show()
```

Output:-

```
Population Mean: 3.9
Mean of Sampling Distribution: 3.9083999999999994
```

